

COMPONENTS

Isolation amplifier

Toshiba has announced the launch of optical-coupled isolation amplifiers with a delta-sigma AD converter that can detect infinitesimal electric current and voltage fluctuation with high precision, for use in industrial equipment applications. These will be available in analogue output and digital output types, and can be provided in either a low-height 2.3mm (maximum) SO8L thin package or a conventional DIP8 package.

Toshiba Corp.

Website: www.toshiba.co

System on chip

Renesas has announced the development of R-Car T2 system on chip (SoC), dedicated to Ethernet AVB-enabled vehicle camera networks, complementing the company's pR-Car family devices for infotainment, instrument cluster and advanced driver assistance system applications. The product enables delivery of high-resolution camera video through multiple systems while maintaining real-time performance with low latency levels.

Renesas Electronics Corp.

Website: www.renesas.com

AC/DC modules

RECOM has launched its latest generation of low-power modules with rated outputs of 2W or 3W. One of the main new features is the extended operating temperature range. As a consequence, power supplies of RAC02-SE/277 and RAC03-SE/277 series can now be used in areas where temperatures drop as low as -40°C.



The modules accommodate a wide input power range of 85V AC to 305V AC (120V DC to 430V DC) and are available with output voltages of 3.3V DC, 5V DC, 12V DC and 24V DC.

RECOM Power GmbH

Website: www.recom-power.com

Charger-interface IC

Power Integrations has announced its newest member of ChiPhy charger-interface integrated circuit (IC)



family, CHY103D, the first IC for off-line AC-DC chargers compatible with Quick Charge 3.0 protocol from Qualcomm Technologies Inc. It includes many protection features including adaptive output over-voltage protection that prevents the output from exceeding 120 per cent of the set output voltage, output soft short-circuit protection that detects partial shorts and stops power delivery to prevent overheating of cable, and connector and a remote shutdown protection feature that enables the powered device to shut the adaptor off remotely if a fault is detected.

Power Integrations

Website: www.power.com

Solid-state relay

Vishay's newly-launched relays are single-pole, single-throw switches.



These feature a modular construction consisting of a gallium aluminium arsenide (GaAlAs) IRED actuation control and MOSFETs for the switch output. These are offered in DIP-6 (VOR1142A6), SMD-6 (VOR1142B6) and SOP-4 (VOR1142M4) packages.

Devices in the SMD-6 and DIP-6 packages offer a high isolation test voltage of 5300V/min, while devices in the smaller SOP-4 offer an isolation test voltage of 3750V/min. All relays offer low typical on-resistance of 22Ω, high load current of 140mA, load voltage of 400V and very low turn-on current of 0.25mA.

Vishay Intertechnology Inc.

Website: www.vishay.com

Power MOSFETs

STMicroelectronics has extended its MDmesh M2 series of N-channel power MOSFETs with a new family of devices that offer high power efficiencies in power supplies for servers, laptops, telecom and consumer applications, especially under light-load conditions.

Key features of the MDmesh M2 series include:

- Extremely-low Qg; as low as 16nC
- Optimised capacitance profile for light-load conditions
- Optimised Vth and Rg values for soft switching
- Body diode ruggedness

STMicroelectronics

Website: www.st.com

Optocoupler

Everlight Electronics has introduced a single-channel 8-pin SOP 3.3V 15MBit/s logic gate optocoupler for use in industrial applications. The EL071L consists of an infrared emitting diode optically coupled to a CMOS detector IC. The high-speed optocoupler EL071L is lead-free, halogen-free and RoHS-compliant.

Other features include both 3.3V and 5V CMOS compatibility, a high CMR of more than 10,000V/μs for higher/superior anti-interference and 10kV/μs minimum common mode transient immunity.

Everlight Electronics Co. Ltd

Website: www.everlight.com